Bureau Veritas, Saudi Arabia

6th Floor, Al Raja Tower, King Fahad Road, Bandriyah, Al Khobar (31952), Saudi Arabia

Tel No : +966 13 882 1071 **Web** : https://middle-east.bureauveritas.com

Report No.: 241030

LIFTING EQUIPMENT INSPECTION REPORT	**Area** DAMMAM	**Inspection Date** 27/09/2024		**Inspection Time** 10:00	**Previous Sticker** No.24-1342	Issue By BV
Department / Contractor SUDHIR EQUIPMENT RENTAL CO.	**Equipment Number** SRC-RT-090-023	**Equipment Location** DMM YARD		**Inspection Result** **Pass**	**New Sticker No.** BV-24-2078	
Manufacturer SANY	**Model** SRC900T	**Capacity** 90 T	**Type** Mobile Crane	**Equipment Serial Number** RC0090CE0556	**Sticker Expiration Date** 26/12/2024	

Above Equipment was inspected by **Bureau Veritas** in accordance with applicable Saudi Aramco GI's, ASME and API Standards. Deficiencies that require corrective action are listed below. Specific repairs to correct each deficiency should be noted in the corrective action taken column.

No.	DEFICIENCIES / NOTES	Corrective Action Taken
1	-PERIODIC INSPECTION AND FUNCTION TEST FOR MOBILE CRANE AS PER AS S.ARAMCO GI 7.030 AND ASME B30.5 -MAIN HOOK BLOCK S/N: SRC-023-01, SWL: 90 TON @ 6 FALLS - AUX HOOK S/N: SRC-023-02, SWL: 6.5 TON @ 1FALL - FLY JIB WAS STOWED	-

Receiver Name : NIHAL		**Signature :**	**Inspector Name :** Mohamed OMARA
Badge No.: -	**Telephone No.:** -		**Telephone :** 055 078 7628

When re-inspection is requested, a complete copy of this report should be ready review by the inspector.

SAUDI ARAMCO INSPECTION CHECKLIST	SAIC NUMBER **SAIC-U-7007**	ISSUE DATE **12-Jan-23**	DISCIPLINE **ELEVATING/LIFTING EQUIPMENT**
MOBILE / CRAWLER CRANES			
PROJECT TITLE	WBS/BI/JO NUMBER	INSPECTION COMPANY **Bureau Veritas, Saudi Arabia**	

EQUIPMENT ID NUMBER :	EQUIPMENT DESCRIPTION/TYPE :		SERIAL NO. :	REPORT No.: 241030
SRC-RT-090-023	Mobile Crane		RC0090CE0556	
CAPACITY (SWL) : 90 T	MANUFACTURER : SANY		FACILITY / LOCATION : DMM YARD	
REQUESTED INSPECTION DATE :	RFI No.	TASK COMPLETED DATE : 27/09/2024	EQUIPMENT OWNER : SUDHIR EQUIPMENT RENTAL CO.	

ACCEPTANCE CRITERIA

1.0 GENERAL REQUIREMENTS

		REFERENCE	PASS	FAIL	NA	REMARKS
1.1	Equipment documentation is available	ASME B30.5	✓			
1.2	Preventive Maintenance is established and updated records are maintained	ASME B30.5 Sec:5-2.3.1	✓			
1.3	Daily pre-operational inspection checklists are checked	GI 7.030 Para: 5.1.2	✓			
1.4	Vehicle government registration card (ISTIMARA) and Government periodic inspection are maintained and valid at all times for equipment that has a vehicle plate.	GI 7.030 Para: 7.1.2 Note:5	✓			
1.5	Durable load chart is provided in an accessible location	ASME B30.5	✓			
1.6	Work areas chart is available	ASME B30.5	✓			
1.7	Operator is certified (for the inspected crane type/model) and licensed	G.I. 7.025	✓			
1.8	A sign is posted warning the operator not to rely solely on any automatic device as a substitute for safe operating practice	G.I. 7.030	✓			
1.9	Crane is painted safety yellow	SAES-B067			✓	

		REFERENCE	PASS	FAIL	NA	REMARKS
1.10	Crane is painted safety yellow-and-black stripes on offshore facilities	SAES-B067			✓	
1.11	Rated capacity of crane is marked (Arabic & English)	SAES-B067	✓			
1.12	Fire extinguisher is in place with minimum rating 10 BC	ASME B30.5	✓			
1.13	Markings and direction of all control levers are clearly displayed	ASME B30.5	✓			
1.14	Crane backward stability is in accordance with manufacturers standard (minimum radius and no load condition)	ASME B30.5	✓			

2.0 INSPECTION POINTS (MOBILE & CRAWLER CRANES)

		REFERENCE	PASS	FAIL	NA	REMARKS
2.1	Load- holding check valve for boom is operational	ASME B30.5	✓			
2.2	Two (2) full wraps of rope remain on the drum when hook is in the extreme low position	ASME B30.5	✓			
2.3	Swing control is smoothly operating on acceleration and deceleration	ASME B30.5	✓			
2.4	Swing braking operates, as well as positive locking devices (manual and auto)	ASME B30.5	✓			
2.5	Travel controls are properly operating/functioning	ASME B30.5	✓			
2.6	Auxiliary travel controls function correctly for multiple control- station cranes	ASME B30.5			✓	
2.7	Travel brakes and locks for crawler cranes are correctly operating	ASME B30.5			✓	
2.8	Travel brakes and locks for wheel-mounted cranes are correctly operating	ASME B30.5	✓			

		REFERENCE	PASS	FAIL	NA	REMARKS
2.9	Power plant controls of superstructure -mounted power plant working properly	ASME B30.5	✓			
2.10	Rotation-resistant and fiber-core ropes are not used for boom hoist reeving	ASME B30.5			✓	
2.11	Sheave grooves have surface smoothness without any defects	ASME B30.5	✓			
2.12	Sheave guards are in place on the lower block	ASME B30.5	✓			
2.13	All sheaves can be lubricated (except permanently lubricated types)	ASME B30.5	✓			
2.14	All cab glazing's are safety glazing material	ASME B30.5	✓			
2.15	Verify cab doors functionality and condition	ASME B30.5	✓			
2.16	Seat belt is fitted and in good condition	ASME B30.5	✓			
2.17	Walking platforms and areas of structure are skid-resistance	ASME B30.5	✓			
2.18	Guard rails are fitted for crawler cranes	ASME B30.5			✓	
2.19	Fixed or telescoping bumper for boom stop is functional (if fitted)	ASME B30.5			✓	
2.20	Shock absorbing bumper for boom stop is functional (if fitted)	ASME B30.5			✓	
2.21	Hydraulic boom elevating cylinder, used as boom stop, is functional (if fitted)	ASME B30.5	✓			
2.22	Boom angle indicator is fitted, operable, and can be read from the operator station	ASME B30.5	✓			
2.23	Boom shut-off or hydraulic relief is provided for high angle boom condition	ASME B30.5	✓			
2.24	Boom length indicator is fitted for telescoping booms and is working	ASME B30.5	✓			

		REFERENCE	PASS	FAIL	NA	REMARKS
2.25	Anti-two-blocking devices are fitted and working (main and auxiliary)	ASME B30.5	✓			
2.26	Operation and accuracy of the fitted LMI (compare with load chart over a range of angles and radii)	ASME B30.5	✓			
2.27	LMI is fitted, operational and accurate for bool length, rated load, angle, radius, etc. (for 3 ton cranes or more)	ASME B30.5 Sec: 5-1.9.10.2	✓			
2.28	Outriggers and jacks are not leaking oil	ASME B30.5	✓			
2.29	Jack is in good operating condition at each outrigger	ASME B30.5	✓			
2.30	Each outrigger jack locking device is functional (if fitted)	ASME B30.5	✓			
2.31	No leakage from each jack seal when the crane is jacked up	ASME B30.5	✓			
2.32	Outrigger jack bolts are fully secure	ASME B30.5	✓			
2.33	All road wheel nuts are fully secure	ASME B30.5	✓			
2.34	All tyres are serviceable and without damage or excess wear	ASME B30.5	✓			
2.35	Fuel tank supports and straps are secure	ASME B30.5	✓			
2.36	Hydraulic oil tank supports and straps are secure	ASME B30.5	✓			
2.37	Underside of chassis shows no obvious faults	ASME B30.5	✓			
2.38	Outrigger boxes are not cracked	ASME B30.5	✓			
2.39	Front and rear steering cylinder anchor points and pins are secure	ASME B30.5	✓			
2.40	Track rod end joints are not worn and are secure	ASME B30.5	✓			
2.41	Axle bolts and nuts are fully secure	ASME B30.5	✓			
2.42	Wheel hub oil seals are not leaking	ASME B30.5	✓			

		REFERENCE	PASS	FAIL	NA	REMARKS
2.43	Brake linings are not oil contaminated	ASME B30.5	✓			
2.44	Brake hoses have not deteriorated, e.g. cracks, splits, leaks, etc.	ASME B30.5	✓			
2.45	All chassis cross members are not cracked or corroded	ASME B30.5	✓			
2.46	Transmission mounting bolts and bushings are secure	ASME B30.5	✓			
2.47	All drive shaft bolts and universal joints are not worn	ASME B30.5	✓			
2.48	Ring gear bolts are fully secure	ASME B30.5	✓			
2.49	All hydraulic hoses under chassis have not deteriorated (cracks , splits , leaks)	ASME B30.5	✓			
2.50	Engine shows no major oil leakage	ASME B30.5	✓			
2.51	Radiator does not leak water	ASME B30.5	✓			
2.52 (a)	Rope is free of damages: 1 broken wire protruding from the core Wear of 1/3 of the original diameter of outside individual wires Kinking, crushing, birdcaging or other distortion	ASME B30.5	✓			
2.52 (b)	For running ropes: Max of 6 randomly broken wires in 1 lay or 3 broken wires in 1 strand of 1 lay, For rotation resistance ropes: 2 randomly distributed broken wires in six rope diameters or 4 randomly distributed broken wires in 30 rope diameters	ASME B30.5	✓			
2.52(c)	Evidence of any heat damage from any cause Reduction below nominal diameter	ASME B30.5	✓			
2.53	Main hoist drum mounting bolts are secure	ASME B30.5	✓			
2.54	Rear telescope side and top wear pads are serviceable	ASME B30.5	✓			
2.55	Bottom and side wear pads of telescope sections are serviceable	ASME B30.5	✓			
2.56	Center swivel coupling pipes and hoses do not leak	ASME B30.5	✓			

		REFERENCE	PASS	FAIL	NA	REMARKS
2.57	Center swivel coupling bolts are all secure	ASME B30.5	✓			
2.58	Boom lift cylinder anchor bolts and bushings are secure	ASME B30.5	✓			
2.59	Swing gearbox mounting bolts are secure	ASME B30.5	✓			
2.60	Swing gearbox bottom oil seal and drive gear are in a serviceable condition (Should be well lubricated)	ASME B30.5	✓			
2.61	Wire rope clips shall be drop-forged steel of the single saddle U bolt or doubled saddle type clip for end termination.	ASME B30.5	✓			
2.62	Emergency switch / Power take off switch need to be verified	ASME B30.5	✓			

3.0 CRANE CAB

		REFERENCE	PASS	FAIL	NA	REMARKS
3.1	Horn and wind speed anemometer are working	ASME B30.5	✓			
3.2	Crane level indicator is fitted and working	ASME B30.5	✓			
3.3	Engine gauges and associated indicators are working	ASME B30.5	✓			
3.4	The warning sign stating power line required clearances is visible to the operator	ASME B30.5	✓			
3.5	Load block sheaves turn freely and are undamaged	ASME B30.5	✓			
3.7	Hook assemblies are equipped with safety latches	ASME B30.5	✓			
3.8	Hook assembly labeling and manufacturer data is clearly marked	ASME B30.10	✓			
3.9	Hook's weight is clearly marked/printed on the hook	ASME B30.10	✓			
3.10	Safe working load of hook is clearly marked on the item	ASME B30.10	✓			

		REFERENCE	PASS	FAIL	NA	REMARKS
3.11	Hook does not show defects such as nicks, cracks and gouges	ASME B30.10	✓			
3.12	Hook is not bent or twisted	ASME B30.10	✓			
3.13	Hook is not distorted in the throat opening - Max. allowable throat opening is 15% compared to new hook, or as per manufacturer recommendations	ASME B30.10	✓			
3.14	Maximum wear in the hook bowl is not exceeding 10% (compared to new hook) or as per manufacturer recommendations	ASME B30.10	✓			
3.15	Hook is not cracked, gouged or shows nicks	ASME B30.10	✓			
3.16	Hook can lock (if it is a self-locking hook)	ASME B30.10			✓	
3.17	Inoperative latch (if provided) — any damaged latch or malfunctioning latch that does not close the hook's throat	ASME B30.10 Sec : 10-1.10.5 (i)	✓			
3.18	Rope guards and locked in position with pins (hook block)	ASME B30.5	✓			
3.19	Dead end rope anchor is fitted with a lock pin (hook block)	ASME B30.5	✓			
3.20	Dead end rope anchor is correctly made with a suitable wedge socket and loop- back termination	ASME B30.5	✓			
3.21	Reverse travel alarm is operational	ASME B30.5	✓			
3.22	Turning signals and indicators are working correctly	ASME B30.5	✓			
3.23	Road lighting, including spotlights, are in good condition and operable	ASME B30.5	✓			
3.24	Dry friction brakes and clutches shall be protected against rain and other liquids such as oil and lubricants	ASME B30.5 Sec 5-1.9.8	✓			

4.0 CRAWLER CRANE

		REFERENCE	PASS	FAIL	NA	REMARKS
4.1	Track pads are undamaged	ASME B30.5			✓	
4.2	Drive sockets and tumblers are serviceable	ASME B30.5			✓	
4.3	Top and bottom swing rollers and tracks are in good overall condition (Should be well lubricated)	ASME B30.5			✓	
4.4	Track idlers are serviceable	ASME B30.5			✓	
4.5	Back boom stops are functioning	ASME B30.5			✓	
4.6	Back A-Frame is in good condition	ASME B30.5			✓	
4.7	Boom cut-out (at maximum angle) is operable	ASME B30.5			✓	
4.8	Power low lowering is operational	ASME B30.5			✓	
4.9	Boom hoist safety auxiliary ratchet / pawl lock is engaging	ASME B30.5 Sec: 5-1.3.1(c)			✓	
4.10	Boom chords and lacings are undamaged (bent)	ASME B30.5			✓	
4.11	Sheave grooves are free from surface defects that could cause rope damage.	ASME B30.5 Sec: 5-1.7.4 (a)			✓	
4.12	Pendant ropes are showing no broken wires (see 2.0.55)	ASME B30.5			✓	
4.13	Operation of steering brakes is satisfactory	ASME B30.5			✓	
4.14	Main machinery guarding is in place	ASME B30.5			✓	
4.15	Main clutch is fully operational	ASME B30.5			✓	
4.16	Main boom and hoist drum brakes are fully functional	ASME B30.5			✓	

REMARKS

-

REFERENCE DOCUMENTS:
1- ASME B30.5 - 2021 Issue: Mobile & Locomotive Cranes
2- ASME B30.10 - 2019 Issue: Hooks
3- SAES-B067 - 2020 Issue: Safety Identification and Safety Colors
4- SA GI 7.025-2018 Issue: Heavy Equipment Operator and Rigger Testing and Certification
5- SA GI 7.030 - 2023 Issue: Inspection and Testing Requirements for Elevating/Lifting Equipment

Signature

Inspected / Witnessed by 3rd Party Certified Inspector

Name	Mohamed OMARA
Signature	
Certificate No.	80019106
Company	Bureau Veritas

User Representative

Name	NIHAL
Signature	
Date	27/09/2024

NO.BV-24-2078

BUREAU VERITAS

Tel.: 013 822 1071
Fax: 013 822 1072

EQUIP NO.: SRC-RT-090-023

EQUIP S. N. : RC0090CE0556

DATE INSPECTED : 27/09/2024

NEXT INSPECTION DATE : 26/12/2024

INSPECTED BY : Mohamed OMARA

SA INSPECTION DEPARTMENT APPROVED CONTRACTOR AGENCIES